

Oluwatimilehin (Timi) Emmanuel Owolabi

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Education

Covenant University

Bachelor of Engineering in Electrical & Electronics Engineering, GPA: 4.85/5.0

Ogun, Nigeria

Sep 2020 – Aug 2025

- **Thesis:** [Coordinated Control of Quadrotor Swarms: Classical, Adaptive, and Learning-Based Methods](#)

Technical Skills

Languages: Python, C++, SQL

ML & Robotics: PyTorch, TensorFlow, JAX, Hugging Face, MLflow, FastAPI, LangChain, ROS2, MuJoCo, CasADi

DevOps & Cloud: Docker, Kubernetes, AWS, GCP, Helm, Terraform, GitHub Actions, Argo CD, Grafana, OpenTelemetry

Professional Experience

Quidax Technologies

Lagos, Nigeria

Platform Engineer (Graduate Trainee)

Jan 2026 – Present

- Optimized AWS EKS autoscaling with Karpenter, increasing CPU utilization from 20% to 50% and memory utilization from 30% to 70%, reducing infrastructure costs by 8% while maintaining reliability.
- Modernized legacy manual Kargo deployment workflows by migrating Node, Python, Go, and Java services to reusable GitHub Actions and Argo CD pipelines, helping achieve 98% automated deployment/failover coverage across 1,099 jobs.
- Created and configured dedicated non-exchange Traefik ingress classes across development, staging, and production, separating non-exchange services from exchange resources and reducing traffic through the core exchange load balancer.
- Enabled mobile telemetry ingestion for the App team by developing and deploying authenticated OTLP proxies that forwarded traces, logs, and metrics to internal OTEL collectors and observability backends.
- Built an isolated sandbox environment for merchants to safely test integrations and workflows without impacting production.
- Instrumented the merchant platform with distributed tracing and logging using OpenTelemetry, enabling engineers to root-cause production incidents independently and reducing cross-team escalations.
- Led the engineering track of QuidaxOS, an AI agent that supports Engineering, Product, and Design teams from product discovery to deployment, reducing handoff friction and aligning execution with company objectives.

Trotta Technologies, Inc

San Francisco, California (Remote)

Founding Machine Learning Engineer

Jan 2026 – Feb 2026

- Founding engineer and ML team lead on a short-term contract, building the company's AI infrastructure on Google Cloud Vertex AI and shipping a production multi-agent defense system for phishing, deepfake audio, and synthetic video detection.
- Developed a self-improving AI threat detection system using LangSmith-hosted agents and anomaly detection neural networks on Google Cloud Run, reducing malicious email and phone call verification time from about 1 hour to under 20 seconds while enabling continuous self-supervised model retraining from production signals.

Scrella Technologies

Lagos, Nigeria

Lead Machine Learning Engineer

Jul 2024 – Dec 2024

- Led a three-person ML team and built the company's AI infrastructure on AWS SageMaker, architecting the full MLOps lifecycle from data collection through production deployment.
- Designed and deployed an AI-powered phone inspection pipeline using fine-tuned YOLOv11, CLaHE, spectral analysis, and Google PaliGemma to detect device imperfections from user videos, eliminating manual approval and reducing onboarding review time from 5 minutes to 18 seconds.

Research Experience

ML Collective

Lagos, Nigeria

Embodied AI Research Group Lead

Jun 2026 – Present

- Achieved an $11\times$ inference speed-up of NVIDIA's Cosmos Policy robot model, as measured by latency falling from 624 ms to 57 ms per action chunk on an A100, with $13.5\times$ fewer FLOPs, $3.2\times$ less GPU memory and a 95.5% success rate on LIBERO benchmark, by cutting 16 of its 28 transformer blocks, replacing 5-step diffusion sampling with a single forward pass, and distilling the full model into a 0.88B-parameter student named SmolWAM (NeurIPS 2026 workshop paper).
- Led a team of 5 on three concurrent robot-learning projects: a Jev-based robot policy (System One decision model), learning from demonstration, and SmolWAM.

WaySense — Dr. Daniel Omeiza, Oxford Robotics Institute

Oxford, United Kingdom (Remote)

Undergraduate Research Assistant

Mar 2024 – Sep 2024

- Analyzed the Lyft Level 5 dataset to estimate multi-agent vehicle positions and velocities, detect lane-change maneuvers, and improve explainability of autonomous driving decisions using ego- and agent-centric representations.

- Pioneered adaptation of the fast subset-scan algorithm on Graph Neural Networks to identify statistically significant anomalies in traffic datasets.

Publications & Manuscripts

T. **Owolabi**, H. Olawore, F. Adelakun, and A. Oyewumi, “SmolWAM: A Latent World-Action Model for Robotic Manipulation on Edge Devices”, *NeurIPS 2026 Robotics World Modeling Workshop*, 2026.

T. E. Somefun, T. **Owolabi**, and O. M. Longe, “[Practical Trade-offs in Neural Network Optimization: Brute Force Search and Gradient Descent](#)”, *Engineering Research Express*, 2025.

A. Awelewa, K. Ojo, T. Abimbola-Oladejo, T. **Owolabi**, “[Fuzzy-PID Controller for Liquid Level Control of Tank Systems](#)”, *NIPES-Journal of Science and Technology*, 2025.

T. E. Somefun et al. and T. **Owolabi**, “[Energy Optimization Algorithm for Reducing Energy Consumption in a Smart Home](#)”, *IEEE ICMEAS*, 2023.

O. Idowu et al. and T. **Owolabi**, “[Enhancing Radiological Imaging for Better Healthcare Outcomes Through High Performance Hybrid Approach](#)”, *ASRIC Journal on Engineering Sciences*, 2024.

Selected Projects	
SmolWAM Python, PyTorch, LIBERO	Sep 2026
• Achieved an $11\times$ inference speed-up of NVIDIA's 2B-parameter Cosmos Policy robot model (624 to 57 ms on an A100, $13.5\times$ fewer FLOPs, $3.2\times$ less memory) while keeping 95.5% LIBERO success, by pruning 16 of 28 transformer blocks, replacing 5-step diffusion with a single forward pass, and distilling it into SmolWAM, a 0.88B student (NeurIPS 2026 Robotics World Modeling Workshop).	
Autonomous Quadcopter C++, ESP32, PlatformIO	2025
• Implemented an ESP32 flight-controller with modular drivers for sensors, receiver input, state estimation, PID control, motor actuation, battery monitoring, EEPROM, and telemetry.	
• Built cascaded attitude/rate PID control, 50 Hz 16-bit ESC PWM motor mixing, GPS/barometer/IMU/compass sensor fusion, PPM receiver decoding, and a ground-station UI.	
Coordinated Control of Multi Quadrotor Swarms Python, PyTorch, JAX, PyBullet	2025
• Modeled quadrotor dynamics in quaternion and Euler state-space form, then implemented PID, LQR with integral action, MRAC, PPO, and MPC controllers for position and trajectory tracking.	
• Developed minimum-snap trajectory planning and multi-agent formation controllers using leader-follower, consensus, behavioral rules, and potential-field collision avoidance in gym-pybullet-drones.	
Pick-and-Place Robot Arm Python, JAX, MuJoCo, Gymnasium, Stable-Baselines3	2025
• Built custom Gymnasium environments around a MuJoCo Franka Panda scene, exposing actuator control ranges and observations from qpos, qvel, cube pose, end-effector pose, and finger poses.	
• Trained staged PPO policies for reaching, grasping, and lifting with reward shaping for distance, grasp quality, contact-based grasp detection, cube height, drop penalties, randomized resets, and success-rate evaluation.	

Awards and Honors	
Finalist, Africa Deep Tech Challenge	Ongoing
First Class Honours, Dept. of Electrical and Information Engineering, Covenant University	2025
Finalist, Pan-African Robotics Competition	2024
Founder's Award, Best WASSCE Result, Wellspring College	2021
Valedictorian, Department of Science, Wellspring College	2020

Selected Talks	
AI: Our Current Reality and Future Trajectory	Nigerian Society of Engineers, August 2024
Deep Learning: Neural Networks	APWEN, April 2024

Teaching, Leadership & Community Impact	
Engineering Innovation Empowerment Foundation (EIEF) Co-Founder	Abuja, Nigeria Dec 2024 – Present
• Designed and produced hands-on engineering kits to teach core engineering principles.	
• Organized and led technical training programs, mentoring students in practical engineering skills.	
• Launched The Classroom Project to provide tutoring, mentoring, and learning materials for low-resource public secondary schools.	
Association of Electrical and Information Engineering Students President	Ogun, Nigeria Sep 2024 – Aug 2025
• Led a 20-member executive team in coordinating projects and competitions for over 2,000 students.	

- Launched a student mentorship program pairing juniors with seniors for academic and career guidance.

Schneider Electric

Lagos, Nigeria

Sustainability Intern

Aug 2023 – Oct 2023

- Partnered with NGOs to deliver STEM tutoring aligned with Schneider Electric’s global initiative to empower 1 million youths.
- Reviewed NGO support budgets exceeding \$60,000 each, ensuring alignment with sustainability program objectives.

Enactus

Ogun, Nigeria

Member (National Champions)

May 2021 – May 2023

- Designed a soil-nutrient sensor circuit for the “Cow Blood to Manure” sustainability project.
- Contributed to the Pet City initiative which uses recycled plastic bottles to make interlocking road tiles.
- Supported the Bargass project which involves converting plant biomass waste into reusable food packs.